



Windows Security Logs Assignment (Core)

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Coding Dojo Cybersecurity Boot Camp
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Windows Security Logs (Core)



Learning Objectives:

- Use open-source tools to solve platform-related issues.
- Analyze Windows Event Logs.

Setting Up

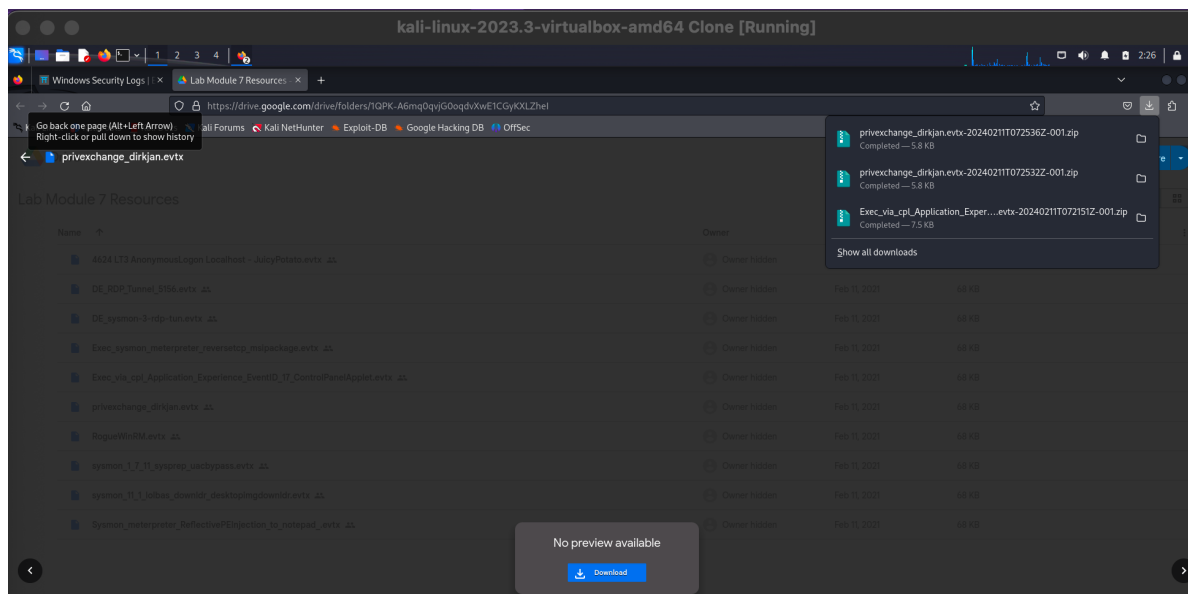
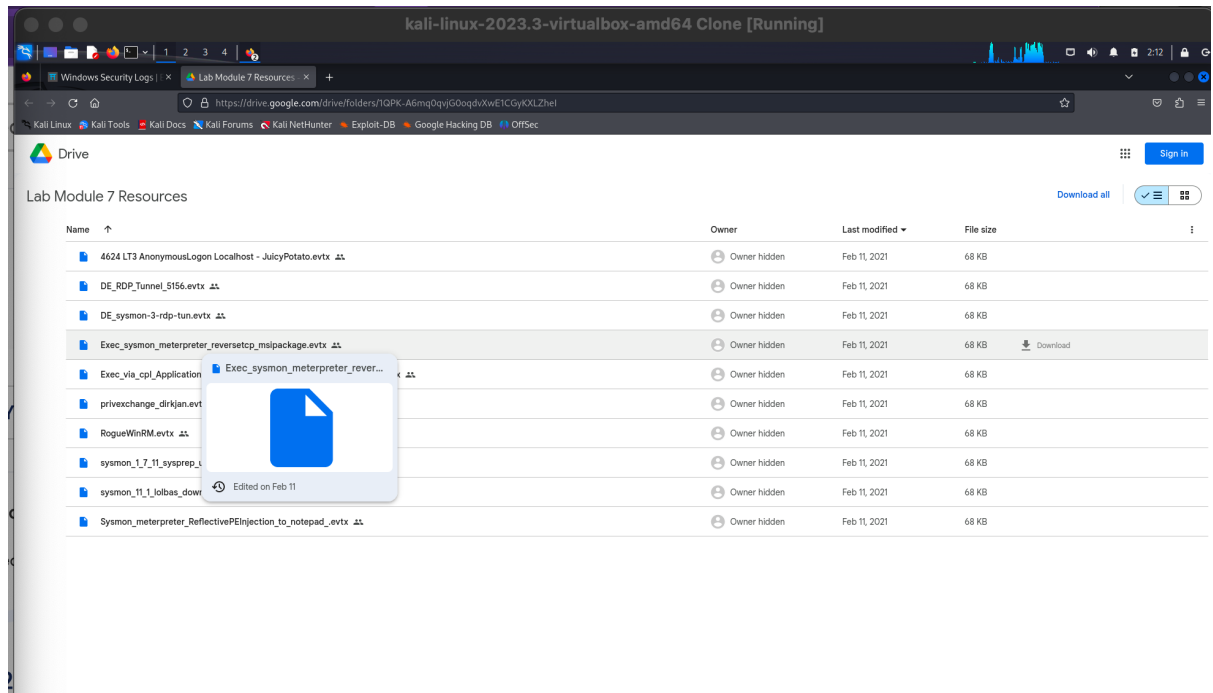
1. Open Kali Linux.
2. Download two files from [this google drive folder](#).
3. Download and use [this tool](#) to be able to read the .evtx files in Linux.
4. For each file that you downloaded onto your machine, give a 1 paragraph narrative summary of the types of activities being performed on the system from the logs.

Assignment Submission

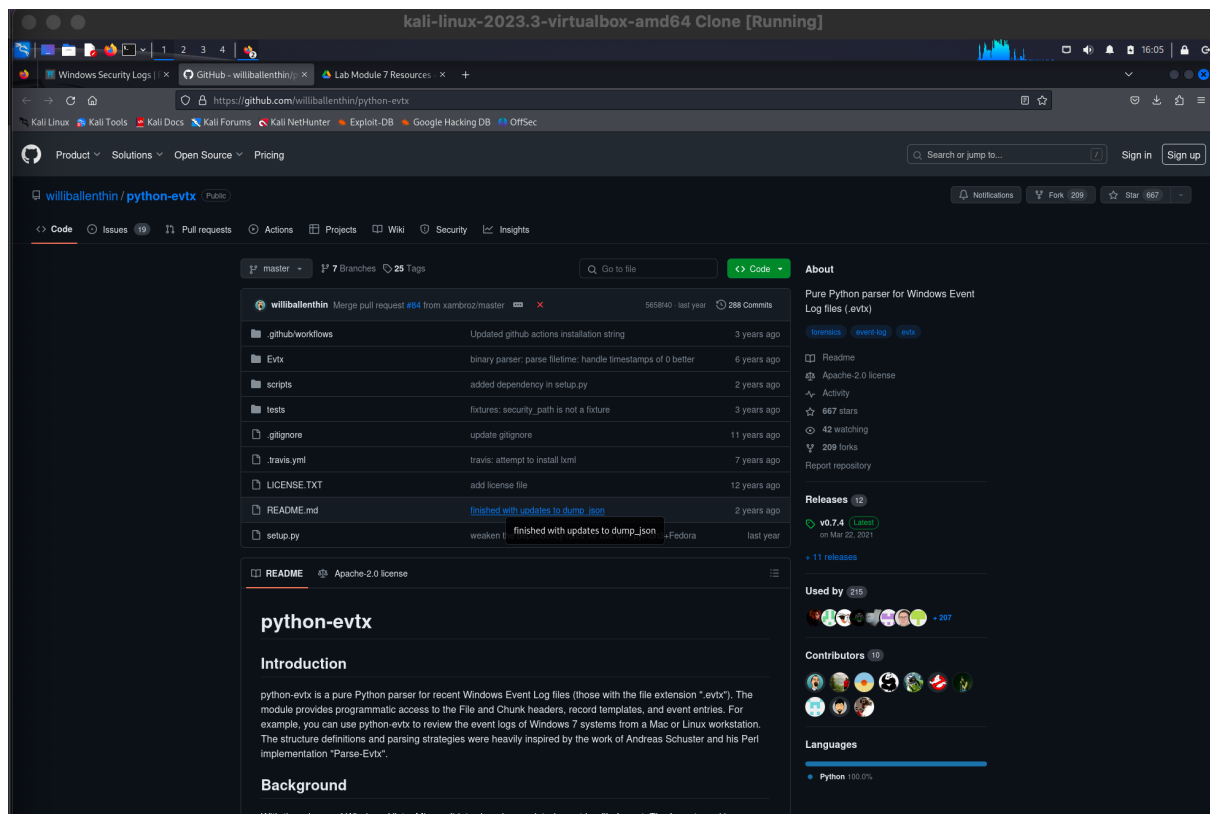
Upload a single document with the narrative summaries detailing which files you chose. Provide screenshots of the logs to back up any claims in your summary.

Assignment Solution:

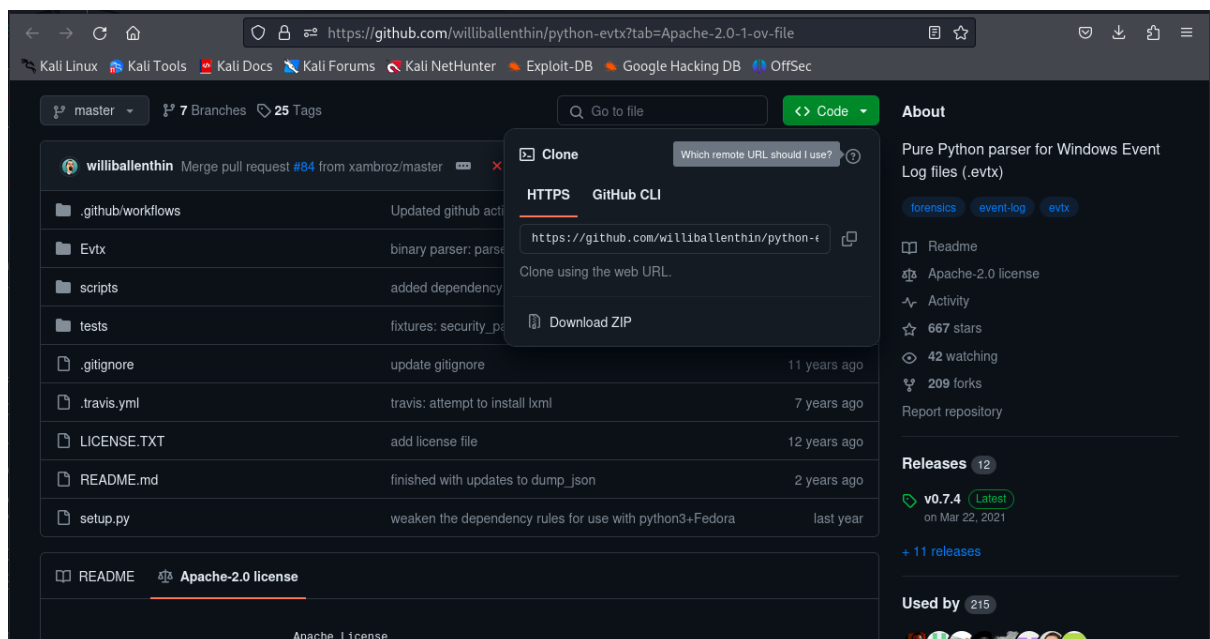
1. Kali Linux opened.
2. 2 files downloaded from [this google drive folder](#).



3. Download and use [this tool](#) to be able to read the .evtx files in Linux.



- Select the 'Code' button:



- Copy the URL or click the 'copy' icon to copy.
- Type command and URL as displayed below:



```
(kali㉿kali)-[~/Downloads]
$ git clone https://github.com/williballenthin/python-evtx.git
```

- Click 'Enter' to execute the command above.

```
(kali㉿kali)-[~/Downloads]
$ git clone https://github.com/williballenthin/python-evtx.git
Cloning into 'python-evtx' ...
remote: Enumerating objects: 1631, done.
remote: Counting objects: 100% (117/117), done.
remote: Compressing objects: 100% (50/50), done.
remote: Total 1631 (delta 62), reused 103 (delta 59), pack-reused 1514
Receiving objects: 100% (1631/1631), 3.20 MiB | 4.08 MiB/s, done.
Resolving deltas: 100% (958/958), done.
```

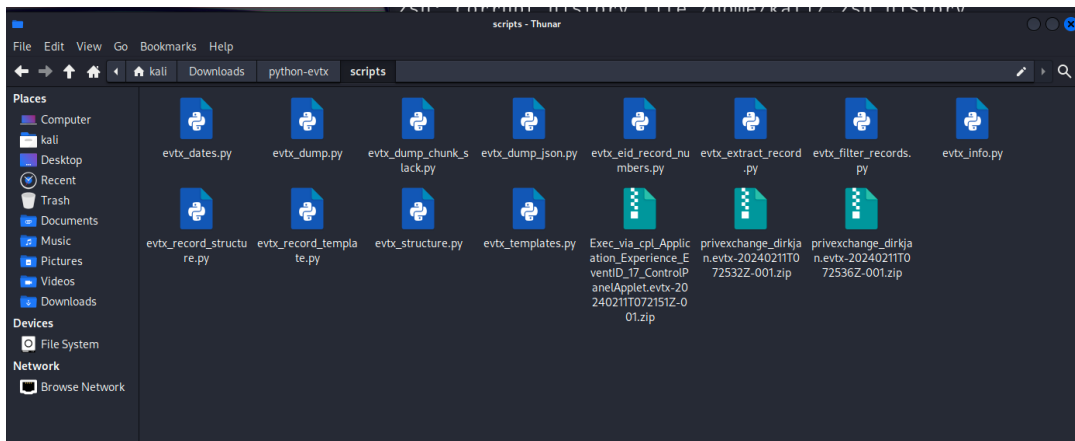
```
(kali㉿kali)-[~/Downloads]
$
```

- Read through the README file. It is recommended to install python-evtx.

```
(kali㉿kali)-[~/Downloads]
$ pip install python-evtx
Defaulting to user installation because normal site-packages is not writeable
Collecting python-evtx
  Downloading python_evtx-0.7.4-py3-none-any.whl (35 kB)
Requirement already satisfied: six in /usr/lib/python3/dist-packages (from python-evtx) (1.16.0)
Collecting hexdump=3.3 (from python-evtx)
  Downloading hexdump-3.3.zip (12 kB)
  Preparing metadata (setup.py) ... done
Collecting more-itertools=5.0.0 (from python-evtx)
  Downloading more_itertools-5.0.0-py3-none-any.whl (52 kB)
    52.3/52.3 kB 1.4 MB/s eta 0:00:00
Requirement already satisfied: zipp=1.0.0 in /usr/lib/python3/dist-packages (from python-evtx) (1.0.0)
Collecting configparser=4.0.2 (from python-evtx)
  Downloading configparser-4.0.2-py2.py3-none-any.whl (22 kB)
Collecting pyparsing=2.4.7 (from python-evtx)
  Downloading pyparsing-2.4.7-py2.py3-none-any.whl (67 kB)
    67.8/67.8 kB 2.7 MB/s eta 0:00:00
Building wheels for collected packages: hexdump
  Building wheel for hexdump (setup.py) ... done
  Created wheel for hexdump: filename=hexdump-3.3-py3-none-any.whl size=8894 sha256=6ecfbbde5490a4c0af29a3c9a1e58c16af08d7e72486afc12a0a08abbaeb4bad
  Stored in directory: /home/kali/.cache/pip/wheels/67/21/60/6c8dd0e145d13fd9501e1078cfd8aa34069f022cf4cdda6b0f
Successfully built hexdump
Installing collected packages: hexdump, pyparsing, more-itertools, configparser, python-evtx
Successfully installed configparser-4.0.2 hexdump-3.3 more-itertools-5.0.0 pyparsing-2.4.7 python-evtx-0.7.4
```

```
(kali㉿kali)-[~/Downloads]
$
```

- Copy the event files into the **scripts** folder of the python-evtx directory.



```
kali@kali: ~/Downloads/python-evtx/scripts
File Actions Edit View Help
└─$ cd python-evtx
cd: no such file or directory: python-evtx

(kali@kali)~[~/Downloads]
└─$ cd python-evtx

(kali@kali)~[~/Downloads/python-evtx]
└─$ ls -lrtl
total 36
-rw-r--r-- 1 kali kali 2035 Feb 11 16:21 setup.py
-rw-r--r-- 1 kali kali 4345 Feb 11 16:21 README.md
-rw-r--r-- 1 kali kali 11358 Feb 11 16:21 LICENSE.TXT
drwxr-xr-x 2 kali kali 4096 Feb 11 16:21 Evtx
drwxr-xr-x 3 kali kali 4096 Feb 11 16:21 tests
drwxr-xr-x 2 kali kali 4096 Feb 11 22:31 scripts

(kali@kali)~[~/Downloads/python-evtx]
└─$ cd scripts

(kali@kali)~[~/Downloads/python-evtx/scripts]
└─$ ls -lrtl
total 216
-rw-r--r-- 1 kali kali 69632 Feb 11 2021 Exec_via_cpl_Application_Experience_EventID_17_ControlPanelApplet.evtx
-rw-r--r-- 1 kali kali 69632 Feb 11 2021 privexchange_dirkjan.evtx
-rw-r--r-- 1 kali kali 7696 Feb 11 02:21 Exec_via_cpl_Application_Experience_EventID_17_ControlPanelApplet.evtx-20240211T072151Z-001.zip
-rw-r--r-- 1 kali kali 5943 Feb 11 02:25 privexchange_dirkjan.evtx-20240211T072532Z-001.zip
-rw-r--r-- 1 kali kali 5943 Feb 11 02:25 privexchange_dirkjan.evtx-20240211T072536Z-001.zip
-rwxr-xr-x 1 kali kali 1547 Feb 11 16:21 evtx_templates.py
-rwxr-xr-x 1 kali kali 6556 Feb 11 16:21 evtx_structure.py
-rwxr-xr-x 1 kali kali 736 Feb 11 16:21 evtx_record_template.py
-rwxr-xr-x 1 kali kali 3324 Feb 11 16:21 evtx_record_structure.py
-rw-r--r-- 1 kali kali 4477 Feb 11 16:21 evtx_info.py
-rwxr-xr-x 1 kali kali 1785 Feb 11 16:21 evtx_filter_records.py
-rwxr-xr-x 1 kali kali 1462 Feb 11 16:21 evtx_extract_record.py
-rwxr-xr-x 1 kali kali 924 Feb 11 16:21 evtx_eid_record_numbers.py
-rwxr-xr-x 1 kali kali 1352 Feb 11 16:21 evtx_dump.py
-rw-r--r-- 1 kali kali 3436 Feb 11 16:21 evtx_dump_json.py
-rwxr-xr-x 1 kali kali 1846 Feb 11 16:21 evtx_dump_chunk_slack.py
-rw-r--r-- 1 kali kali 1801 Feb 11 16:21 evtx_dates.py

(kali@kali)~[~/Downloads/python-evtx/scripts]
└─$
```

- Use the evtx_dump.py python script to parse the event files one after the other:

```
(kali@kali)-[~/Downloads/python-evtx/scripts]
$ python evtx_dump.py Exec_via_cpl_Application_Experience_EventID_17_ControlPanelApplet.evtx
```

For each file that you downloaded onto your machine, give a 1 paragraph narrative summary of the types of activities being performed on the system from the logs.

1. Exec_via_cpl_Application_Experience_EventID_17_ControlPanelApplet.evtx

```
File Actions Edit View Help
$ python evtx_dump.py Exec_via_cpl_Application_Experience_EventID_17_ControlPanelApplet.evtx
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>

<Events>
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Microsoft-Windows-Program-Compatibility-Assistant" Guid="{4cb314df-c11f-47d7-9c04-65fb0051561b}"></Provider>
<EventID Qualifiers="">17</EventID>
<Version>0</Version>
<Level>4</Level>
<Task>0</Task>
<Opcode>0</Opcode>
<Keywords>0x4000000000000000</Keywords>
<TimeCreated SystemTime="2019-07-18 21:50:20.049400"></TimeCreated>
<EventRecordID>1</EventRecordID>
<Correlation ActivityID="" RelatedActivityID=""></Correlation>
<Execution ProcessID="5020" ThreadID="7468"></Execution>
<Channel>Microsoft-Windows-Application-Experience/Program-Compatibility-Assistant</Channel>
<Computer>MSEDGWIN10</Computer>
<Security UserID="S-1-5-18"></Security>
</System>
<UserData><ResolverFiredEvent xmlns="http://www.microsoft.com/Windows/Diagnosis/PCA/events"><ExePath>
C:\Users\IEUser\Desktop\procxp64.exe</ExePath>
<ResolverName>WrpKeyMitigation</ResolverName>
</ResolverFiredEvent>
</UserData>
</Event>

<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Microsoft-Windows-Program-Compatibility-Assistant" Guid="{4cb314df-c11f-47d7-9c04-65fb0051561b}"></Provider>
<EventID Qualifiers="">17</EventID>
<Version>0</Version>
<Level>4</Level>
<Task>0</Task>
<Opcode>0</Opcode>
<Keywords>0x4000000000000000</Keywords>
<TimeCreated SystemTime="2019-07-29 21:11:17.282188"></TimeCreated>
<EventRecordID>2</EventRecordID>
<Correlation ActivityID="" RelatedActivityID=""></Correlation>
<Execution ProcessID="4396" ThreadID="3944"></Execution>
<Channel>Microsoft-Windows-Application-Experience/Program-Compatibility-Assistant</Channel>
<Computer>MSEDGWIN10</Computer>
<Security UserID="S-1-5-18"></Security>
</System>
<UserData><ResolverFiredEvent xmlns="http://www.microsoft.com/Windows/Diagnosis/PCA/events"><ExePath>
```

The event logs indicate a series of occurrences related to program compatibility assessment on the system named "MSEDGEWIN10". These assessments were triggered by the execution of different programs, including "procexp64.exe" and "Invoice@0582.cpl", located in various directories on the system. The Program Compatibility Assistant detected potential compatibility issues with these programs and attempted to mitigate them using resolutions such as "WrpKeyMitigation" and "ControlPanelApplet". These logs reflect the system's proactive monitoring and management of software compatibility to ensure smooth operation and user experience.

2. privexchange_dirkjan.evtx

```
(kali@kali)-[~/Downloads/python-evtx/scripts]
$ python evtx_dump.py privexchange_dirkjan.evtx
```

- Screenshot of excerpts of the log is displayed below:

```
(kali@kali)-[~/Downloads/python-evtx/scripts]
$ python evtx_dump.py privexchange_dirkjan.evtx
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>

<Events>
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"></Provider>
<EventID Qualifiers="">4776</EventID>
<Version>0</Version>
<Level>0</Level>
<Task>14336</Task>
<Opcode>0</Opcode>
<Keywords>0x8020000000000000</Keywords>
<TimeCreated SystemTime="2019-02-02 09:16:52.479004"></TimeCreated>
<EventRecordID>65966</EventRecordID>
<Correlation ActivityID="" RelatedActivityID=""></Correlation>
<Execution ProcessID="612" ThreadID="1280"></Execution>
<Channel>Security</Channel>
<Computer>ICORP-DC.internal.corp</Computer>
<Security UserID=""></Security>
</System>
<EventData><Data Name="PackageName">MICROSOFT_AUTHENTICATION_PACKAGE_V1_0</Data>
<Data Name="TargetUserName">helpdesk</Data>
<Data Name="Workstation">evil.internal.corp</Data>
<Data Name="Status">0x00000000</Data>
</EventData>
</Event>

<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"></Provider>
<EventID Qualifiers="">4624</EventID>
<Version>2</Version>
<Level>0</Level>
<Task>12544</Task>
<Opcode>0</Opcode>
<Keywords>0x8020000000000000</Keywords>
<TimeCreated SystemTime="2019-02-02 09:16:58.787262"></TimeCreated>
<EventRecordID>65967</EventRecordID>
<Correlation ActivityID="" RelatedActivityID=""></Correlation>
<Execution ProcessID="612" ThreadID="1884"></Execution>
<Channel>Security</Channel>
<Computer>ICORP-DC.internal.corp</Computer>
<Security UserID=""></Security>
</System>
<EventData><Data Name="SubjectUserSid">S-1-0-0</Data>
<Data Name="SubjectUserName">-</Data>
```

The provided event logs indicate various security-related activities on the system "ICORP-DC.internal.corp". Firstly, two successful authentication events occurred, one involving the user "helpdesk" authenticating from the workstation "evil.internal.corp" using the Microsoft authentication package

"MICROSOFT_AUTHENTICATION_PACKAGE_V1_0", and another involving the system account "EXCHANGE\$" authenticating from the workstation "EXCHANGE" using the Kerberos authentication package. Additionally, an object access event was logged, indicating that the user "EXCHANGE\$" accessed a directory service object. Finally, another successful authentication event occurred, this time involving the system account "ICORP-DC\$" authenticating via Kerberos from an unspecified workstation. These logs collectively reflect user and system authentication activities, as well as object access monitoring, contributing to the overall security posture of the system.